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Harmonizing Subsurface Data Platforms for Agentic AI in Upstream: A Dual Layer Architecture Using OSDU and Microsoft Fabric

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Abstract

Even in the present day and age, the upstream oil and gas industry continues to face significant challenges in integrating siloed subsurface data across domains such as seismic, wells, reservoirs and production. As part of **ENERGYai** – an industrial AI initiative advancing upstream intelligence – we present a dual-layer data architecture that harmonizes two platforms: the Open Subsurface Data Universe (OSDU) as a canonical subsurface data layer, and Microsoft Fabric as a semantic and analytics layer. This integrated architecture creates a robust foundation for **agentic AI** in oil & gas upstream workflows. We outline how diverse data types (seismic surveys, well logs, petrophysical measurements, production records, etc.) are unified in OSDU, then virtualized into Fabric's Lakehouse environment via OneLake.

This approach retains a single source of truth while enabling rich semantic models and cross-domain analytics without duplicating data. We also describe specific AI agents (for metadata tagging, document parsing, QA/QC, and ingestion planning) that capitalize on this architecture to automate data management tasks. Key outcomes include improved data quality and lineage, faster data ingestion throughput, and enhanced user access to data. Ultimately, geoscientists and engineers are empowered to spend less time on data wrangling and more on value-added analysis, demonstrating clear efficiency of gains and business value.

Introduction

For the Oil and Gas Industry, Upstream exploration and production (**E&P**) undeniably involves extremely complex subsurface workflows. These workflows start from seismic acquisition and interpretation to drilling, production, and reservoir management. Each of these steps in this lifecycle generates huge volumes of different data types (3D/2D seismic volumes, well logs, core samples, images, reservoir models, production time-series, engineering reports), which are often managed by different teams using specialized applications. All these datasets end up being stored in disconnected silos, making it a challenging task to find → integrate → analyze data across disciplines. Industry studies performed within companies and across industries, both have highlighted that geoscientists and data managers spend a substantial portion of their time collecting and preparing data rather than analyzing it [3, 4]. This fragmentation ends up introducing

delays in projects along with hindering cross-domain collaboration, collectively resulting in cloudy and often delayed critical decisions. In short, upstream companies face a data technology dilemma: Huge size + multiple different formats + specialized tools + disconnected journey on disparate systems.

To address this, the industry has been gravitating toward unified data platforms. The Open Subsurface Data Universe (**OSDU**) emerged as a collaborative industry-standard platform to break down these silos. OSDU provides a common data model and API framework for all subsurface and wells data, decoupling data from specific applications. It is envisioned as a single system of record for E&P data, liberating information from application-specific repositories. By enforcing standard schemas and metadata, OSDU promises a **single source of truth** for subsurface data across the enterprise. This aligns with the goals of digital transformation programs (such as the ENERGYai initiative) which seek to leverage AI and analytics on unified datasets to drive exploration efficiency and innovation. ENERGYai's vision of combining data and AI in upstream can greatly benefit from the solid data foundation that OSDU provides.

However, simply centralizing data is not enough, it must also be **analytics ready**. Raw subsurface datasets often need further contextualization, quality control, and combination with business data to extract insights. This is where a complementary platform like **Microsoft Fabric** adds value. Microsoft Fabric is a unified cloud analytics platform (encompassing data lake storage, warehousing, and AI tools) that can layer on top of OSDU. By linking OSDU's curated data into Fabric's Lakehouse and semantic models, we gain powerful capabilities for data virtualization, cross-domain analysis, and AI model development, all while respecting OSDU's governance. In essence, OSDU acts as the stable canonical data layer, and Fabric acts as the agile semantic and AI layer. Together, they form a dual-platform architecture tailored to upstream data needs. Upstream subsurface data is generated at vastly different paces (from multi-year seismic studies to real-time production sensor streams), underscoring the need for a unified platform to manage context and lineage across this broad value chain.

Upstream data lifecycle from exploration to production shown in figure 1. Each phase (exploration, drilling & wells, production, and interpretation/analysis) generates different data types – for example, seismic surveys in exploration, well logs and core samples during drilling, real-time sensor and production data during operations, and models or reports during post-processing. Historically, these datasets were isolated in separate silos, but a unified data platform approach aims to accommodate the full breadth of data across the lifecycle.

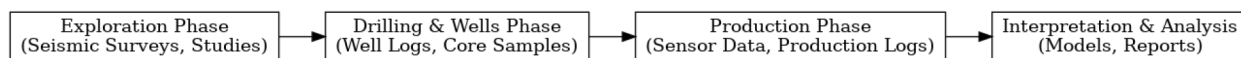


Figure 1—High Level Upstream Data Lifecycle

The remainder of this paper details how a dual OSDU+Fabric architecture can harmonize subsurface data management and enable **agentic AI** – essentially, AI "agents" that automate and augment data engineering tasks. We discuss the data types and integration strategy, present the architecture (with conceptual and technical views), and describe how AI agents are embedded into the workflow. Key innovations of our approach are highlighted, along with challenges encountered (and resolutions) in mapping legacy systems (e.g., Schlumberger Petrel, Techlog, OSIssoft PI) to this modern architecture. We then share the outcomes and benefits observed, and conclude with future directions, including the integration of real-time data streams and an OSDU-compliant agent framework. This blend of technical depth and business value demonstrates a practical innovation that delivers ROI in upstream operations

Methodology and Architecture Data Integration Strategy

In the face of upstream data's complexity and scale, we recognized that no single platform could solve every challenge. To address this, we leveraged the **best of both worlds or divide and conquer strategies**:

OSDU for subsurface data management + lifecycle operations and Microsoft Fabric for everything else which primarily is data analytics, serving layer for metadata captured on process level (as shown in [figure2](#)). Our dual-layer platform utilizes OSDU as the authoritative **system of record** for all subsurface data, while Microsoft Fabric serves as the agile semantic and analytics layer.



Figure 2—Best Of Both World Approach

We began by ingesting data from legacy systems such as the commercial platforms already used in the industry for storing and maintaining the subsurface data. (for master data management). Each dataset was mapped to OSDU's standard schemas and loaded via OSDU's ingestion services into cloud storage. Upon ingestion, each item is registered in OSDU with a unique identifier (UUID/URI) and its metadata is indexed for search and retrieval. This process ensures that previously siloed subsurface datasets are logically unified in one open platform under OSDU's governance.

We also implemented a **unified master data model** to connect related data across domains. For example, entities like *Wells*, *Wellbores*, *Fields*, *Reservoirs*, and *Organizations* are treated as master reference data within OSDU. A well log curve record, for instance, is linked to a specific **Wellbore** entity, which in turn connects to a master **Well** record. This master record has a unique ID that serves as a federated key across all systems, and we carry this key into the Fabric layer as well. By maintaining these master identifiers in both OSDU and Fabric, we enable seamless cross-domain querying and data integration. In practice, this means we can easily join a well's production time-series with its corresponding well information or seismic interpretation using a shared Well identifier. Even if one dataset originates from a PI historian system and another from a seismic database, they become logically linked once ingested into OSDU and mirrored in Fabric. This **federated master key linking** is crucial for building a consistent semantic layer on top of OSDU, directly resolving legacy identity conflicts. By reconciling alias naming and duplicate records, we ensure that each real-world entity (such as a given well or field) is represented by a single master record.

The data ingestion process includes detailed **metadata extraction** for each datatype- capturing attributes such as survey details for a seismic volume on the lines of survey type, processing stage, processing company, size of file, name of file etc. Similarly for well logs, we capture well name, field, file format, source system, acquisition date, units, etc. This approach of capturing metadata of the data file in question along with the logging the details of the process/pipeline/workflow which works on the data file is fundamental for making data **searchable** and **discoverable** based on rich context [5]. Every work product component which ends up being created in OSDU as a part of the process is linked to its relevant master data entities, an example would be wellbores being linked to their respective wells, and every SEG Y volume being linked to the Survey. OSDU ensures the grouping of related data class (and possible data formats) in well-defined domain categories, like well related data will be grouped in WellBore DDMS. In cases where a particular data type did not have complete coverage in OSDU's standard schema, we map the data to the closest available OSDU type – ensuring that every record could be represented in OSDU format without information

loss. For example, we standardized terminology across sources (e.g., using a consistent "GammaRay" log curve name in place of various aliases like "GR") and using FileDDMS to link related data which cannot be added directly to some defined DDMS.

Dual-Layer Architecture Design

Once data is ingested and standardized in OSDU, it is **virtualized** into Microsoft Fabric without the need for physical duplication. **Azure Data Manager for Energy (ADME)** – Microsoft's managed OSDU service – provides connectors that bridge the two platforms, enabling seamless retrieval of OSDU-managed data within Fabric's environment. In our implementation, Fabric acts as a semantic and analytics overlay on top of OSDU's canonical layer, allowing us to query and analyze OSDU data directly from Fabric.

Using Microsoft Fabric's OneLake as the underlying storage abstraction, we implemented **OneLake shortcuts** that reference OSDU's data in place. These shortcuts act as pointers to data managed in OSDU, allowing Fabric to access that data virtually (in-situ) without moving or copying it. For instance, a seismic dataset stored as Parquet files in OSDU's cloud storage can be exposed in Fabric as a Delta Lake table, enabling fast querying through Fabric's engines. This architecture avoids data duplication (reducing storage costs) and eliminates synchronization issues that typically arise when data is copied between systems.

Importantly, OSDU's governance policies — including entitlements, versioning, and audit trails — continue to apply when data is accessed via Fabric. We built custom synchronization processes to ensure that Fabric users can only access data they are authorized to view under OSDU's rules, maintaining consistent security across both layers. In practice, OSDU's access control lists and entitlements are propagated into Fabric, so the **same security model** is enforced regardless of whether a user is pulling data through OSDU or via Fabric's tools.

Integrated dual-layer architecture combining OSDU and Microsoft Fabric. This schematic (figure 3) shows OSDU as the canonical data layer (right, blue) and Microsoft Fabric as the semantic & analytics layer (left, yellow). Legacy data sources are ingested into OSDU's storage via pipelines. OSDU core services (Storage, Search, Delivery, Entitlements) handle data management and standardization across subsurface domains. Microsoft Fabric connects to OSDU data through Azure Data Manager for Energy connectors (dashed arrow), virtualizing the OSDU data into Fabric's OneLake Lakehouse without copying. Data domains not yet supported by OSDU are ingested directly into Fabric as an interim measure (top-left). A unified semantic model & catalog in Fabric overlays both layers, capturing relationships and business context across all data. The consumption layer (bottom) allows users and AI agents to query the data via SQL, GraphQL, or even natural language, and also to retrieve unstructured documents and metadata for AI-driven analysis. Dotted arrows indicate bi-directional data flows: Fabric outputs (derived insights or curated data) can be pushed back into OSDU, creating a feedback loop that continuously enriches the OSDU dataset

On the OSDU side, the ingestion pipelines bring data from various sources into OSDU's storage, and OSDU's core services (Storage, Search, Delivery, Entitlements, etc.) handle the heavy duty of data management – standardizing and securing the data for use across various subsurface domains (wells, seismic, logs, interpretations, etc.). On the Fabric side, the ADME-Fabric connector links directly to the OSDU data store, allowing Fabric to read and analyze the data **in place** without duplicating it. We also ingested certain **ungraduated domain data** – i.e. data from specialized domains not yet fully supported by OSDU's schemas – directly into Fabric as an interim solution. Fabric's Lakehouse can thus combine data referenced via connectors from OSDU with interim domain data that lies outside OSDU's current scope. The semantic model and catalog in Fabric overlay across the two layers, capturing business relationships and definitions for all data. Finally, a consumption layer allows both users and AI applications to interact with the integrated data through standard query interfaces (SQL queries, GraphQL APIs, and even natural

language queries via BI tools), while also enabling retrieval of unstructured documents and metadata for AI-driven analyses [6, 7].

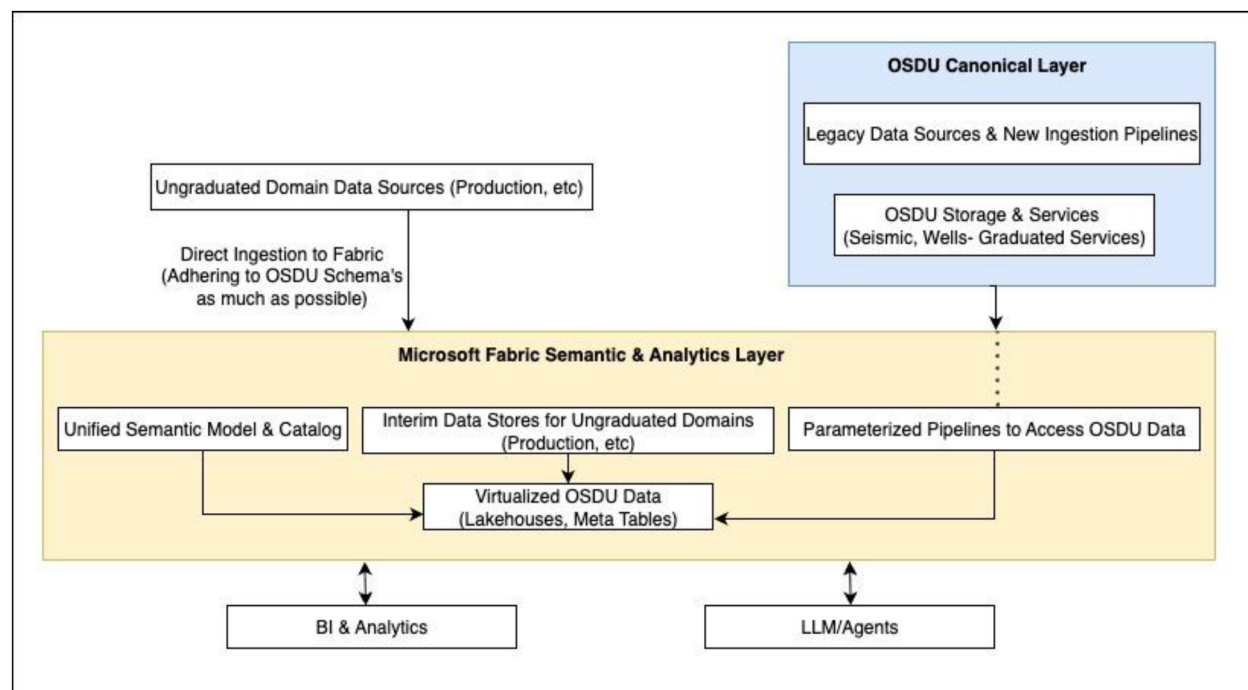


Figure 3—Dual Layer Architecture of Consumption from OSDU & Fabric

In this architecture, OSDU serves as the **single source of truth** for all ingested subsurface data, providing lifecycle management capabilities like data management and governance. And Microsoft Fabric helps in creating advanced analytics layer to serve the AI capabilities while still respecting OSDU's governance rules. Through this architecture, what we have achieved is that any data viewed in Fabric is always traceable back to its OSDU record of origin, and updates or interpretation generated through the analytics layer of Fabric can be ingested back into OSDU. This creates a feedback loop that improves and enriches the master dataset over time. The result is a unified, cohesive system for data management, discovery, and analytics across the subsurface domain.

Integration of AI Agents in the Workflow

A key innovation in our approach is embedding **AI agents** at various points in the data workflow to automate and accelerate data management tasks. Four specialized agents were developed, each targeting a pain point in the data lifecycle:

- **Ingestion Planning Agent:** This agent intelligently orchestrates the data ingestion process. It analyzes incoming data sources and backlogs to prioritize what to ingest next, scheduling jobs for optimal throughput. For streaming data (e.g. daily production sensor feeds), the agent ensures they are ingested promptly (near real-time), whereas large bulk data backlogs are queued and processed in parallel in the background. By dynamically scheduling and balancing loads, this agent maximizes ingestion velocity and ensures new data becomes available as quickly as possible.
- **Document Parsing Agent:** Many subsurface datasets come in unstructured forms (reports, PDFs, scanned documents). We deployed an AI document parser that uses natural language processing to read and extract relevant information from documents during ingestion. For example, if a well report PDF is uploaded, the parsing agent can extract metadata (well name, coordinates,

key results) and even digitize tables or figures. The extracted content is then fed into OSDU as structured metadata or linked documents. This saves manual effort and enriches the dataset – making previously opaque documents **queryable**.

- **Metadata Tagging Agent:** Once a record is ingested, this agent automatically augments and standardizes its metadata. It uses rule-based and ML-driven techniques to tag records with missing attributes (e.g., adding a coordinate reference system if missing, or tagging a seismic survey with its basin and field based on text content). It also ensures metadata consistency – for instance, normalizing units of measure or enforcing naming conventions. The tagging agent dramatically improved metadata completeness (approaching 100% filled fields for all records) and ensures new data is immediately **discoverable** via search.
- **QA/QC Agent:** We have broadly divided the realm of Quality checks/controls into two categories: Technical & Business QA/QC. The technical qa/qc agent is an inherent part of every data ingestion which applies the defined checks to different data classes before they can be qualified for ingestion. It scans for anomalies such as specific data standards mismatches (e.g., textual header missing important information such as incorrect byte locations for inlines/crosslines in the case of Seismic volumes. The agents also as a byproduct of the process, create a pdf report describing the detail of all the checks performed on the incoming file and their results with reasoning. It then proceeds to provide a recommendation on whether to ingest this data or to send it for verification to a data manager.

The diagram in figure 4 illustrates how each AI agent contributes in a traditional data pipeline: an **ingestion planning agent** prioritizes & schedules incoming data loads, a **document parsing agent** extracts structured information from unstructured files during the ingestion process, a **metadata tagging agent** enriches each ingested record with standardized metadata attributes, and a **QA/QC agent** validates data quality and consistency before loading.

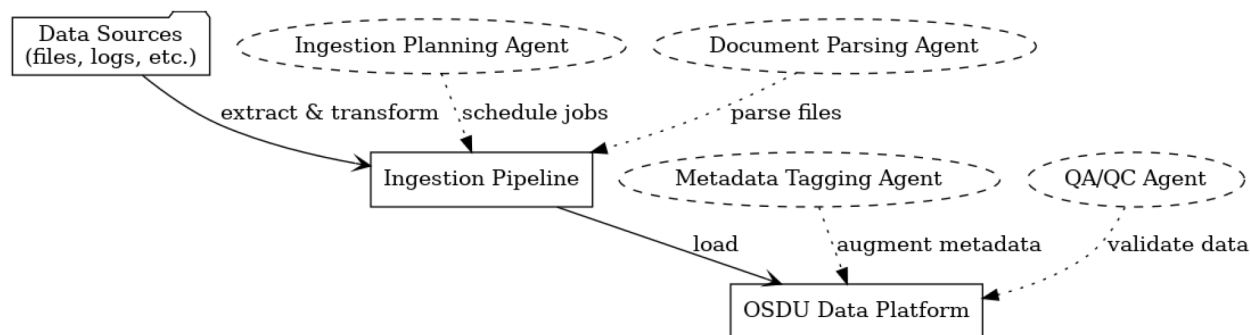


Figure 4—Agentic enabled pipelines

By integrating these AI agents, our data platform became an **active, self-improving system** rather than a passive data repository. The agents operate as tireless digital colleagues to our data managers – handling tedious, repetitive tasks at scale and ensuring that data quality is maintained even as volumes grow. In summary, the dual-layer architecture combined with embedded AI agents allows data management in upstream to keep up with the scale and speed of modern operations, with minimal human intervention.

Next Steps and Future Directions

During implementation, we encountered certain specialized data domains that were not fully supported by OSDU's schemas at the time. For instance, **production data** (daily production volumes, real-time well rates), **Rock and Fluid Sample (RAFS)** laboratory analysis results, and some advanced **reservoir simulation outputs** did not yet have complete OSDU data definitions. We managed these datasets within

Fabric as an interim solution. However, we kept them **OSDU-compliant** by defining Fabric Lakehouse tables using the same schemas and attributes that OSDU's standard would use. This means that once the OSDU platform offers full native support for these domains, the data can be migrated into OSDU with minimal effort. In practice, we essentially "pre-modeled" these domain datasets according to OSDU conventions (units, entity relationships, metadata keys), even though they resided in Fabric initially. This foresight ensures a smooth transition in the future, preserving our single-source-of-truth principle. As the OSDU schema expands to cover more domains, we plan to graduate these interim datasets into OSDU's domain services, further reducing reliance on any custom Fabric-only data.

Results and Observations

After implementing the dual-layer OSDU+Fabric platform and the suite of AI agents, we observed significant improvements in data management performance, quality, and user productivity. Below we highlight the key outcomes and benefits, backed by metrics and examples:

- **Faster Data Ingestion and Availability:** The platform dramatically accelerated the speed at which new data becomes available for use. By using parallelized pipelines and avoiding unnecessary data copying, large datasets that previously took months to ingest were loaded in a matter of weeks or days. In one deployment, a critical set of well and seismic data was migrated to OSDU in under a month, whereas earlier estimates for a traditional approach were several months. Certain data types saw major throughput gains – for example, after onboarding the newly created workflows of ingestion with embedded agents, we were able to see reduction of data loading times of up to 2 times. This means that now the required data for better planning is available earlier than expected. Moreover, the **ingestion planning agent** ensured that streaming data (like daily production feeds) were ingested promptly, keeping near-real-time dashboards up to date, while bulk backlogs were processed in the background. Overall, data latency from source to availability was cut drastically, enabling more agile decision-making.
- **Improved Data Quality and Consistency:** The combination of enforced standards in OSDU and AI-driven quality control resulted in markedly cleaner, more trustworthy data. All ingested records now have a comprehensive set of metadata tags (owner, acquisition date, units, quality flags, etc.), approaching 100% metadata completeness. In the legacy silos, many records lacked key descriptors (for instance, ~45% of wells in one legacy database were missing coordinate reference system info); after ingestion and automated tagging, that dropped to 0% – every well in the new platform has complete location metadata. The QA/QC agent caught and helped correct numerous issues early in the process. For example, unit inconsistencies and misalignments that previously went unnoticed (such as a dataset recorded in feet being mistaken for meters) were identified and fixed before they could propagate. The system also reconciled naming differences that had caused duplicate entries – e.g., what were once two "different" wells in separate systems were merged into one master record with aliases. Thanks to these measures, users can now pull data without spending hours on manual cleanup. In practical terms, this means far fewer errors in downstream analyses – engineers no longer stumble over hidden unit conversion mistakes or missing context that could skew results. The data quality improvements have built greater trust in the data across the organization.
- **User Empowerment and Productivity:** By breaking down data silos and providing a one-stop platform, we empowered geoscientists, petrophysicists, and engineers to find and use data much more efficiently. In the past, a geoscientist might spend days or weeks gathering well logs or seismic surveys from various departments and loading them into their tool – now they can discover and retrieve that data within minutes via a unified search or BI portal. Internal user surveys indicate substantial time savings. One asset team reported that the proportion of project time spent on data gathering and formatting dropped from ~60% to ~20% after adopting the new platform, meaning

they now devote 80% of their time to analysis and interpretation (a complete reversal of the prior situation). This aligns with observations noted by industry cloud providers that professionals can "spend less time on data preparation, more on interpretation which our experience confirms [2]. Furthermore, cross-disciplinary collaboration improved as data became more accessible: for instance, a reservoir engineer can easily access geophysical data or well logs through the platform, without needing special conversions or assistance from a custodian. This ability to self-serve data (within proper security bounds) has fostered more integrated studies and faster iterative workflows. The learning curve for new users was eased by the semantic layer and AI assistants – even those unfamiliar with OSDU's APIs could query the data through natural language or predefined dashboards. Overall, the workforce is now able to focus more on *geoscience* and less on data wrangling.

- **AI-Readiness and New Insights:** One of the strategic benefits of having a unified, high-quality dataset is that the organization's subsurface data is now truly *AI-ready*. Data scientists and domain experts can run advanced analytics and machine learning across the entire data corpus – something not feasible when data is fragmented. We have already seen emerging use cases: for example, we trained a machine learning model on thousands of wells' logs from the platform to predict missing log curves, promising accuracy. In another case, we built a recommendation model that suggests relevant analog wells for a given target well (drawing on patterns in the combined data and documents). These kinds of analyses, which leverage the breadth of data, were very difficult before. The platform also enables real-time analytics by combining static and dynamic data: we created prototype *digital twin* dashboards where live production sensor data (streamed into the interim Fabric store) is correlated with historical models and interpretations from OSDU, allowing engineers to get timely alerts and insights. Essentially, the dual-layer architecture provides a solid foundation for EnergyAI initiatives – with all data in one fabric, one can apply AI algorithms to everything from seismic interpretation to drilling optimization. We anticipate that as more data (like real-time drilling or IoT sensors) is integrated, the platform will support closed-loop, AI-driven operations (e.g., an AI agent monitoring real-time drilling parameters and comparing with historical analogs to suggest adjustments on the fly). While full ROI is still being measured, an **Accenture analysis** [1] suggests that such a data+AI platform can improve return on capital employed by on the order of 5-15 percentage points through efficiency gains and better decisions. Our early results – reduced cycle times and more insights per project – indicate substantial value creation is underway.
- **Operational Efficiency and Cost Savings:** From an IT perspective, maintaining one integrated platform has proven more efficient than patchwork of siloed systems. We expect a lower total cost of ownership in the long term, as we can decommission numerous legacy databases and storage silos, consolidating infrastructure and support. The cloud-native, serverless design of Fabric (and the managed OSDU services) means we can scale resources on-demand and pay only for what we use, avoiding the need to provision each team with separate hardware or redundant storage. Additionally, by **ingesting once and reusing data many times**, we eliminate the proliferation of duplicate datasets that used to occur when each group kept their own copies. This not only saves storage and maintenance overhead but also reduces the risk of inconsistencies. The unified entitlement model simplifies data security administration – we manage permissions in one place (OSDU) and it applies everywhere, rather than maintaining ACLs in multiple systems. An unexpected benefit was in external data exchanges: when bringing on a new joint-venture partner or sharing data with an external study team, instead of spending weeks gathering and transferring files, we can now simply grant the external group controlled access to our OSDU/Fabric environment for the relevant data assets. This ease of controlled data sharing (while preserving security and auditability) has accelerated collaboration with partners. Lastly, the platform's automation via

AI agents has reduced manual workload for data managers. For example, the metadata tagging and QA/QC agents together perform what might amount to the work of several full-time data technicians – allowing us to scale up data volume without a linear increase in support staff.

Beyond these quantifiable benefits, there have been positive **cultural shifts** in the organization. Data is increasingly treated as a shared enterprise asset rather than domain-specific property. The visibility of data quality metrics (like completeness scores and quality flags on dashboards) has instilled greater accountability and motivation to "get the data right" at the source. Teams have become more open to adopting AI tools after seeing the success of our agents in handling tedious tasks. In fact, data management has evolved from a low-profile IT chore to a recognized core business enabler in the eyes of leadership, given its clear impact on project timelines and success. In summary, deploying this harmonized subsurface data platform has yielded faster access to data, better data quality, empowered users, and a far greater readiness for advanced digital solutions. It not only solves many of today's pain points but also sets the stage for future capabilities like real-time AI advisory systems and fully data-driven operations. The outcomes we've observed validate the significant business value of unifying subsurface data and AI in a single fabric.

Lessons Learned

Our implementation provided several **lessons learned** that may benefit others undertaking similar projects. First, we found that investing effort upfront in **master data alignment** (resolving naming conflicts and establishing unique IDs for wells, fields, etc.) was worthwhile – it prevented numerous issues down the line and enabled the cross-platform data lineage we needed. Likewise, careful **schema mapping and extension** ensured that legacy data could fit into OSDU without information loss. We learned to leverage OSDU's flexibility (using custom properties where needed) but also to enforce consistency by mapping synonyms to standard terms. Building custom logics for certain legacy workflows required effort, but those components became reusable assets that greatly sped up ingestion of those data sources in subsequent projects.

Performance tuning was also important at scale: we had to optimize search indices and pipeline parallelism to handle the volume (billions of records) efficiently. We found that using Fabric's robust compute engines for heavy queries (e.g., Spark for scanning big files) relieved load on OSDU's services, achieving a good balance. Perhaps the biggest lesson is that technology alone isn't enough – **change management and user buy-in are critical**. We engaged end-users early, demonstrated quick wins (for example, instantly finding a dataset or an AI agent catching a data mistake that users cared about), and iteratively improved the system based on feedback. These steps helped build trust in the platform and enthusiasm for new ways of working.

Conclusions

The convergence of OSDU and Microsoft Fabric in a dual-layer architecture has proven to be a powerful approach to tame upstream data complexity and enable advanced AI-driven workflows. In this paper, we demonstrated how harmonizing subsurface data platforms addresses the long-standing fragmentation problem: OSDU provides the robust, standardized data foundation, while Fabric adds a flexible, intelligent analytics layer on top. This layered architecture retains a **single source of truth** for all subsurface data yet makes that data immediately usable for insights through virtualization and semantic modeling. By integrating the two platforms, we achieved what neither alone could fully deliver – OSDU's rigorous data management **plus** Fabric's rich analytics and AI capabilities – functioning together as one ecosystem. The incorporation of **agentic AI** further elevates the platform from a passive repository of information to an active, self-improving system. We deployed AI agents for metadata tagging, document processing, quality control, and ingestion management, effectively creating an environment where data handling can keep up with the scale and speed of modern exploration projects. These agents act as tireless digital assistants to our

data team, ensuring that data on the platform remains comprehensive, consistent, and ready for consumption. The result is a unified data fabric that not only consolidates data but also continuously curates and enriches it.

The outcomes observed validate the benefits of this approach. We achieved measurable improvements in data accessibility, quality, and user productivity, which collectively de-risk and accelerate upstream decision-making. Geoscientists can now derive insights more quickly from a complete data picture, asset teams can collaborate on a unified subsurface view, and leadership can trust that decisions are based on the full wealth of available information rather than a fragmented subset. In an industry where companies strive to "do more with less" and push the boundaries of automation (as exemplified by initiatives like ENERGYai), our dual-platform solution offers a blueprint for maximizing the value of data.

It shows that by embracing open standards, cloud scalability, and AI-driven automation together, significant efficiency gains and innovation are attainable in upstream workflows. In our case, the platform modernized data infrastructure and delivered tangible business value – from faster project cycles to new analytic insights – making a strong case for broader adoption of such architectures in the energy sector.

Looking forward, there are several exciting directions to extend this work. **First** is the development of an **OSDU-compliant agent framework**. While our AI agents were implemented in an ad-hoc manner, a future evolution could formalize how AI services plug into OSDU – for example, defining standard event hooks so that whenever new data is ingested, it triggers an AI workflow automatically, or creating a standardized way to store agent-generated metadata and lineage information back in OSDU. We envision that the OSDU community could incorporate an "AI extension" that allows operators to deploy recommended agents (for quality control, metadata enrichment, etc.) as first-class components of the platform. Our implementation can serve as a reference pattern for this, and we intend to share our learnings in forums like The Open Group's OSDU Forum. Such a framework would enable different companies to reuse and share agent "recipes" or models, accelerating AI adoption industry-wide.

Another important future direction is the integration of real-time IoT and streaming data. Our current platform handles mostly static or batch datasets (with some near-real-time feeds from systems like OSIsoft PI). The next step will be to bring high-frequency streaming data from rigs, sensors, and operational sources directly into the data fabric. OSDU is expanding its support for time-series data, and Microsoft Fabric provides streaming analytics capabilities, so technically this is becoming feasible. By incorporating real-time drilling data from sensors, we can enable faster analytics, which further enables faster and accurate decisions. This analytics & IOT convergence where we combine different sources of data which comes from different frequencies could unlock further optimizations & efficiencies. Our architecture is well positioned for this, as it already emphasizes scalability and integration. The main challenges will be handling the sheer volume and velocity of IoT data and ensuring our data quality processes keep up in real-time, but those are areas where cloud-native streaming tech and AI (for anomaly detection, automatic scaling) can help.

Beyond subsurface, the architecture is extensible to other parts of the energy value chain. OSDU is already exploring support for renewable energy data and other domains, and Fabric can easily connect to diverse data sources. This opens the possibility of a **unified data fabric across upstream, midstream, and downstream** operations. For example, subsurface data could be linked with facilities data or even market and financial data to enable cross-domain analytics (such as correlating reservoir characteristics with production economics or with carbon intensity metrics). While our focus was upstream, the principles of a canonical data layer plus a semantic/AI layer can be applied more broadly. We foresee that future digital transformation efforts in energy will increasingly adopt such **data mesh** architectures – federating specialized data platforms (like OSDU for subsurface, potentially analogous ones for wells, facilities, etc.) under a common analytics fabric. Our work is a step in that direction, demonstrating the value of harmonizing data and AI at scale. We believe that as more organizations adopt similar approaches, and as the community of practice grows, the pace of innovation in the industry will significantly increase. Upstream professionals working side-by-side with AI agents in a unified data environment could become the new normal, leading to faster discoveries, safer operations, and more informed, proactive decision-making.

In conclusion, the dual-layer OSDU + Fabric platform with embedded AI agents has shown that it is possible to transform subsurface data management – turning a once fragmented, labor-intensive process into an integrated, automated, and AI-enriched system. The positive results we achieved in data quality, efficiency, and user adoption highlight the considerable value of this approach. We encourage the broader community to build upon these ideas, collaborate on open standards (like OSDU), and share best practices for AI integration. The challenges of the future – whether maximizing recovery, minimizing environmental impact, or optimizing operations – will be best met with **unified data and AI-driven solutions**. The work presented here provides a blueprint for such solutions, and we look forward to its continued evolution and adoption in the industry.

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